AIN SHAMS UNIVERSITY
FACULTY OF ENGINEERING

CSE241 Object-Oriented Programming

HOTEL RESERVATION SYSTEM

Milestone 1

Project Report

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Date: 26/4/2026

1. Introduction

The objective of this project is to design and implement a comprehensive desktop-based system for managing hotel reservations. The system facilitates interactions between three primary users: Guests, Receptionists, and Admins. It addresses the need for a centralized platform to manage room availability, handle secure user registrations, process bookings, and maintain an organized database of amenities and room types.

2. System Overview

The system is built using core Object-Oriented Programming principles to ensure scalability and maintainability.

2.1 OOP Principles Applied

- **Encapsulation:** All model classes (Guest, Staff, Room. etc) use private fields with public getters and setters to protect data integrity.
- **Inheritance**: A common Staff parent class is used to derive Admin and Reception subclasses, sharing attributes like username and password while allowing for specialized behaviors.
- **Polymorphism:** The system identifies staff roles at login using the instanceof operator to provide role-specific dashboards
- **Abstraction**: The Staff class is defined as abstract, ensuring that only specific roles (Admin/Receptionist) can be instantiated.

2.2 Core Classes and Models

- **Guest:** Handles user registration, login, and preferences. It includes validation logic for age (minimum 21 years) and unique usernames.

- **Staff (Admin & Reception): Admin:** Possesses full CRUD (Create, Read, Update, Delete) permissions for rooms, amenities, and room types ,
  Receptionist: Manages the check-in and check-out processes for guests.

- **Room & RoomType:** Models the physical rooms and their categories (Single, Double, Suite).

- **Amenity:** Represents individual services like Wi-Fi, TV, and Mini-bar.

- **Database:** Acts as the application's in-memory data store using static ArrayList fields to maintain persistence during the application's lifecycle.

3. Validation & Exceptions

3.1 Validation Logic

The system implements robust validation to ensure data quality:

- **Age Restriction:** Both Guests and Staff must be born in 2005 or earlier (minimum 21 years old).
- **Unique Usernames:** The system checks the Database during registration to prevent duplicate usernames for both staff and guests.
- **Secure Passwords:** Guest passwords must be at least 8 characters long and include uppercase, lowercase, numbers, and special characters

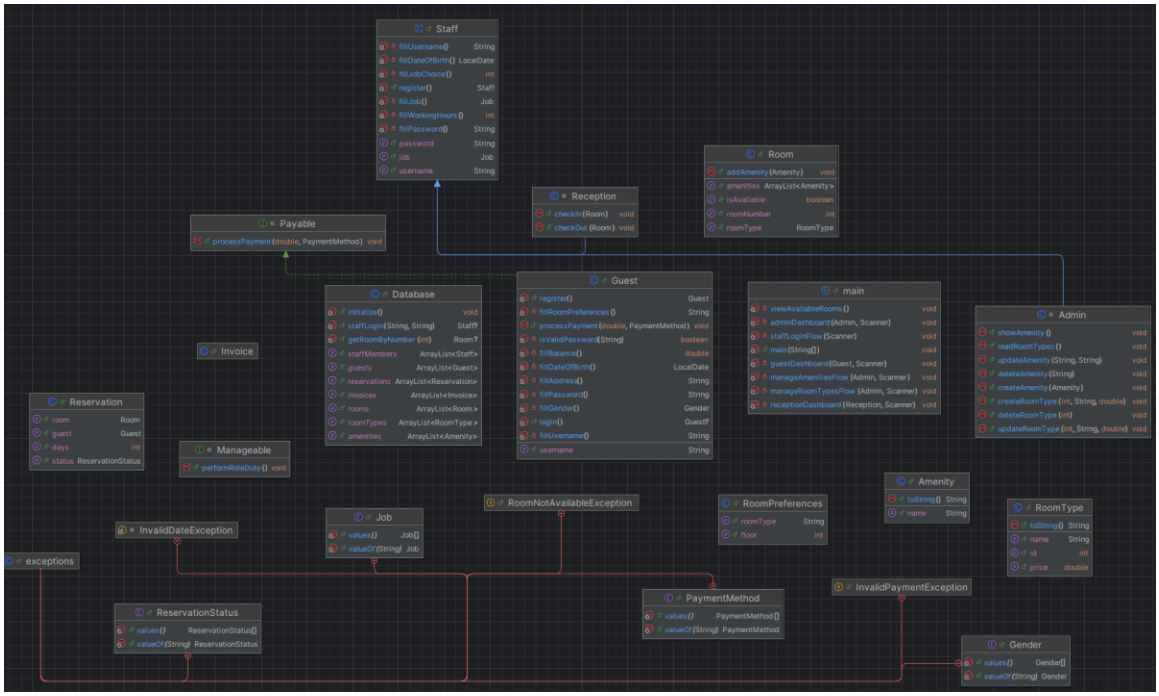
3.2 In-Memory Database

The Database class is pre-populated with dummy data (e.g., Staff: Saif and Ellie; Guests: Ahmed and Sara) to allow for immediate testing and demonstration.

3.3 Exception Handling

Custom exceptions, such as RoomNotAvailableException and InvalidPaymentException, are defined to handle specific business logic errors gracefully.

4. UML Class Diagram



5. Limitations

- **Persistence:** Since the system uses an in-memory database, data is lost once the application is closed.
- **User Interface:** The system currently operates via a command-line interface and doesn't have a user interface.

6. Sample Output

```
C:\Users\hazem\.jdks\openjdk-25.0.2\bin\java.exe "-javaagent:C:\Program Files\JetBrains\IntelliJ IDEA 2025.3.2\lib\idea_rt.jar=63773" -Dfile.encoding=UTF-8 -Dsun.stdout.encoding=UTF-8 -Dsun.st
User 1: Ahmed | Pass: Ahmed@2026
User 2: Sara | Pass: Sara@2026

Staff 1: Saif | Pass: Admin@2026
Staff 2: Ellie | Pass: Staff@2026


 Welcome to the Hotel Reservation System
1. Guest Register
2. Guest Login
3. Staff Register
4. Staff Login
5. Exit
Choice: 5
```

```
  2. Guest Login
  3. Staff Register
  4. Staff Login
  5. Exit
  Choice: 1
  Enter your username: John
  Enter your password: John@2026
  Enter birth year: 2004
  Enter birth month (1-12): 10
  Enter birth day (1-31): 12
  Please enter your balance: 500
  Enter your country:
  egypt
  Enter your city:
  cairo
  Enter your street name:
  orkeid
  Enter your gender (MALE/FEMALE):
  male
  Enter your room preferences: single
  Registration successful for John!
```

```
 --- GUEST DASHBOARD ---
1. View Rooms   2. Book Room   3. My Reservation
4. Cancel       5. Pay Bill    6. Logout
7. SHOW MENU AGAIN

Enter Choice (7 for menu): 3

--- Your Reservations ---
Room: 101 | Status: CONFIRMED | Duration: 3 days
```

```
 Welcome to the Hotel Reservation System
1. Guest Register
2. Guest Login
3. Staff Register
4. Staff Login
5. Exit
Choice: 2
Enter username: John
Enter password: John@2026
Login successful!

 --- GUEST DASHBOARD ---
1. View Rooms  2. Book Room  3. My Reservation
4. Cancel      5. Pay Bill   6. Logout
7. SHOW MENU AGAIN

Enter Choice (7 for menu): 1

 Available Rooms
Room 101 [Single]
   Price: $100.0/night
   Amenities: High-Speed Wi-Fi, Air Conditioning
-----------------------
Room 102 [Double]
   Price: $150.0/night
   Amenities: High-Speed Wi-Fi, 4K Smart TV, Air Conditioning
-----------------------
Room 201 [Suite]
   Price: $300.0/night
   Amenities: High-Speed Wi-Fi, 4K Smart TV, Mini Bar, Jacuzzi Tub, Private Balcony
```

```
Enter Choice (7 for menu): 7

 --- GUEST DASHBOARD ---
1. View Rooms  2. Book Room  3. My Reservation
4. Cancel      5. Pay Bill   6. Logout
7. SHOW MENU AGAIN

Enter Choice (7 for menu): 2

 Available Rooms
Room 101 [Single]
   Price: $100.0/night
   Amenities: High-Speed Wi-Fi, Air Conditioning
-----------------------
Room 102 [Double]
   Price: $150.0/night
   Amenities: High-Speed Wi-Fi, 4K Smart TV, Air Conditioning
-----------------------
Room 201 [Suite]
   Price: $300.0/night
   Amenities: High-Speed Wi-Fi, 4K Smart TV, Mini Bar, Jacuzzi Tub, Private Balcony
-----------------------
Room #: 101
Days: 3
[Success] Room 101 booked for 3 days.
```

```
1. Guest Register
2. Guest Login
3. Staff Register
4. Staff Login
5. Exit
Choice: 4

Select Login Portal:
1. Admin
2. Receptionist
Choice: 1
Staff Username: Saif
Staff Password: Admin@2026

Welcome to the Admin Portal, Saif

 --- ADMIN  CONTROL ---
1. Manage Amenities
2. Manage Room Types
3. Logout
Select Category: 1

 --- AMENITY MANAGEMENT ---
1. List All  2. Add New  3. Update  4. Remove  5. Back
Action: 1

-- Global Amenities --
- High-Speed Wi-Fi
- 4K Smart TV
- Mini Bar
- Air Conditioning
- Private Balcony
- Jacuzzi Tub

 --- AMENITY MANAGEMENT ---
1. List All  2. Add New  3. Update  4. Remove  5. Back
Action:
```

```
 --- ADMIN  CONTROL ---
1. Manage Amenities
2. Manage Room Types
3. Logout
Select Category: 3
Logging out Admin...

 Welcome to the Hotel Reservation System
1. Guest Register
2. Guest Login
3. Staff Register
4. Staff Login
5. Exit
Choice: 4

Select Login Portal:
1. Admin
2. Receptionist
Choice: 2
Staff Username: Ellie
Staff Password: Staff@2026

Welcome to the Receptionist Portal, Ellie

 --- RECEPTION DASHBOARD ---
1. Check-In
2. Check-Out
3. Logout
Choice: 1
Room Number: 101
Room: 101 checked in.

 --- RECEPTION DASHBOARD ---
1. Check-In
2. Check-Out
3. Logout
Choice:
```

7. Design Decisions

- **Role Selection at Login:** We decided to implement a role-selection prompt during the staff login flow. This adds a layer of security by verifying that the user is attempting to access the portal corresponding to their specific Staff type (e.g., an Admin cannot log into the Receptionist portal).

- **Date Management:** We utilized the java.time.LocalDate API for birth dates to handle leap years and date comparisons.

8. Team Contributions

Hazem : Database Class, code integration , main initialization and code testing

Mariam : Guest and Staff classes

Rovan : Reservation and invoice classes

Khaled : Room ,Roomtypes and Amenities Classes

9. References

- Daniel Liang, Introduction to Java Programming, 10th Edition, Pearson, 5 2014.
- Java Documentations

Github repo link :
https://github.com/zoma2077/Hotel-Reservation-System-CSE241.git